

ALGORITHM
Spring 2023, Final Exam, June 1
5:00-8:00 pm

1. (20%) Please answer the following questions:
 - (a) Illustrate the operation of RADIX-SORT on the following numbers: 457, 654, 578, 254, 957, 499, 558, 911, 731, 536.
 - (b) Show that if a node in a binary search tree has two children, then its successor has no left child.
 - (c) Use induction to prove that radix sort works. Where does your proof need the assumption that the intermediate sort is stable?
 - (d) Consider inserting the keys 7, 27, 19, 15, 27, 23, 11, 26 into a hash table of length $m = 13$ using open addressing with the auxiliary hash function $h'(k) = k \bmod 13$. Show the results after the insertion of each number using the linear hashing.
2. (20%) A red-black tree is a balanced binary tree that has height $O(\log n)$.
 - (a) Show the red-black trees that result after successively inserting the keys 5, 22, 13, 15, 27, 33, 11, 21 into an initially empty red-black tree.
 - (b) What is the largest possible number of internal nodes in a red-black tree with black height k ? What is the smallest possible number?
3. (20%) Medians and Order Statistics.
 - (a) Design an algorithm to find both the minimum and the maximum of a set of n elements with $3\lfloor n/2 \rfloor$ comparisons in the worst case.
 - (b) In the algorithm SELECT, the input elements are divided into groups of 5. Will the algorithm work in linear time if they are divided into groups of 7?
4. (20%) Minimizing lateness problem.
 - Single resource processes one job at a time.
 - Job i requires t_i units of processing time and is due at time d_i .
 - If i starts at time s_i , it finishes at time $f_i = s_i + t_i$.
 - Lateness: $\ell_i = \max\{0, f_i - d_i\}$.
 - Goal: schedule all jobs to minimize maximum lateness $L = \max\{\ell_i \mid 1 \leq i \leq n\}$.

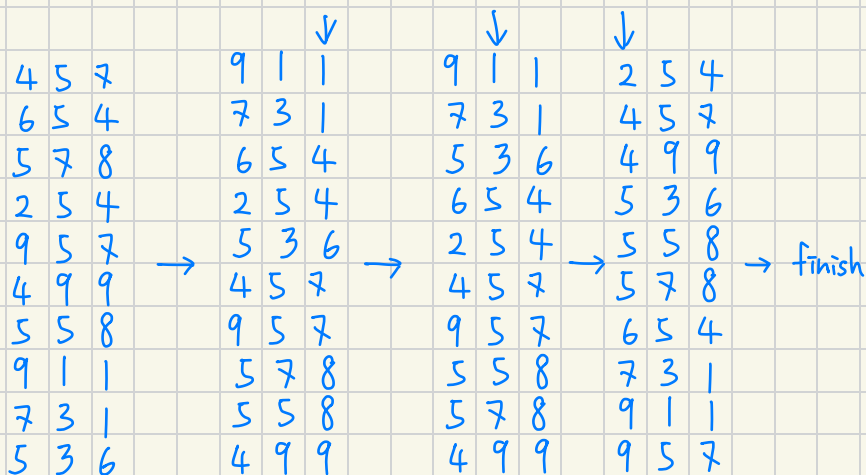
i	1	2	3	4	5	6	7
t_i	3	6	4	1	2	7	5
d_i	10	12	9	8	14	13	11

5. (20%) Determine the cost and structure of an optimal binary search tree for a set of $n = 5$ keys with the following probabilities:

i	0	1	2	3	4	5
p_i		0.15	0.05	0.10	0.10	0.20
q_i	0.10	0.05	0.05	0.05	0.05	0.10

6. (20%) Consider the longest increasing subsequence problem defined as follows. Given a list of numbers $a_1, \dots, a_n$ an increasing subsequence is a list of indices $i_1, \dots, i_k \in \{1, \dots, n\}$ such that $i_1 < i_2 < \dots < i_k$ and $a_{i_1} \leq a_{i_2} \leq \dots \leq a_{i_k}$. The longest increasing subsequence is the longest list of indices with this property.
- What is the longest increasing subsequence of the list 5, 3, 4, 8, 7, 10?
 - Consider the greedy algorithm that chooses the first element of the list, and then repeatedly chooses the next element that is larger. Is this a correct algorithm? Either prove its correctness or provide a counter example.
 - Consider a divide and conquer strategy that splits the list into the first half and second half, recursively computes $L = (\ell_1, \dots, \ell_{k_L})$, $R = (r_1, \dots, r_{k_r})$ the longest increasing subsequences in each half, and then, if the last chosen element in the first half is less than the first chosen index in the second half (i.e. $a_{\ell_{k_L}} \leq a_{r_1}$) returns $L \cdot R$, otherwise it returns the longer of L and R . Is this a correct algorithm? either prove its correctness or provide a counterexample.
 - Design a dynamic programming algorithm for longest increasing subsequence. Prove its correctness and analyze its running time.

1.
(a)



(b)

Let x is a BST nodes have 2 Children.

$\therefore x$ 必有一棵右子樹。

根據 BST 性質，所有 $> x$ 的 node 都在 x 的右子

Assume x 's successor y have left child z

根據 BST 性質，若 z 是 y 的左子，則 $z < y$

而 z 又在 x 的右子樹中，表示 $z > x$

綜上所述，可知關係式 $x < z < y$ 。若存在 1 node z 介於 $x \sim y$ 之間，那麼 z 應該比 y 更接近 x ， y 便不會是 x 的 Successor

→ 這與假設 " y 是 x 的 Successor" 產生矛盾，原假設不成立

→ 若 node x 有 2 Child，其 Successor 必不存在 left Child

(c)

Assume 所有數字皆為 d 位數。

命題 $P(i)$: 經過 i 回合排序後，數字已根據「後 i 個位數」正確排序。

Base Case $i=1$:

演算法針對最低位（第一位）進行排序，完成後，數字顯然已針對最後 1 位數排序，

$P(1)$ 成立。

I.H.: 假設 $i=k-1$ 時成立，即數字會根據「後 $k-1$ 個位數」正確排序。

Inductive Step $i=k$

對 k 位進行排序時，Consider 2 element a, b

→ 若第 k 位位數不同：排序後位置由第 k 位決定，兩者大小關係必正確。

→ 若第 k 位位數相同：其實際大小取決於後 $k-1$ 個位數，由 I.H.，排序前他們已根據「後 $k-1$ 個位數」正確排序，此時必需使用 Stable Sort，才能在第 k 位相同時，保持他們原先的相對順序。 $\Rightarrow P(k)$ 成立

根據數學歸納法，When $i=d$ ， $P(d)$ 成立，全體數字依 d 個位數完全正確排序。

<從後排回來的反例> 512, 253, 218

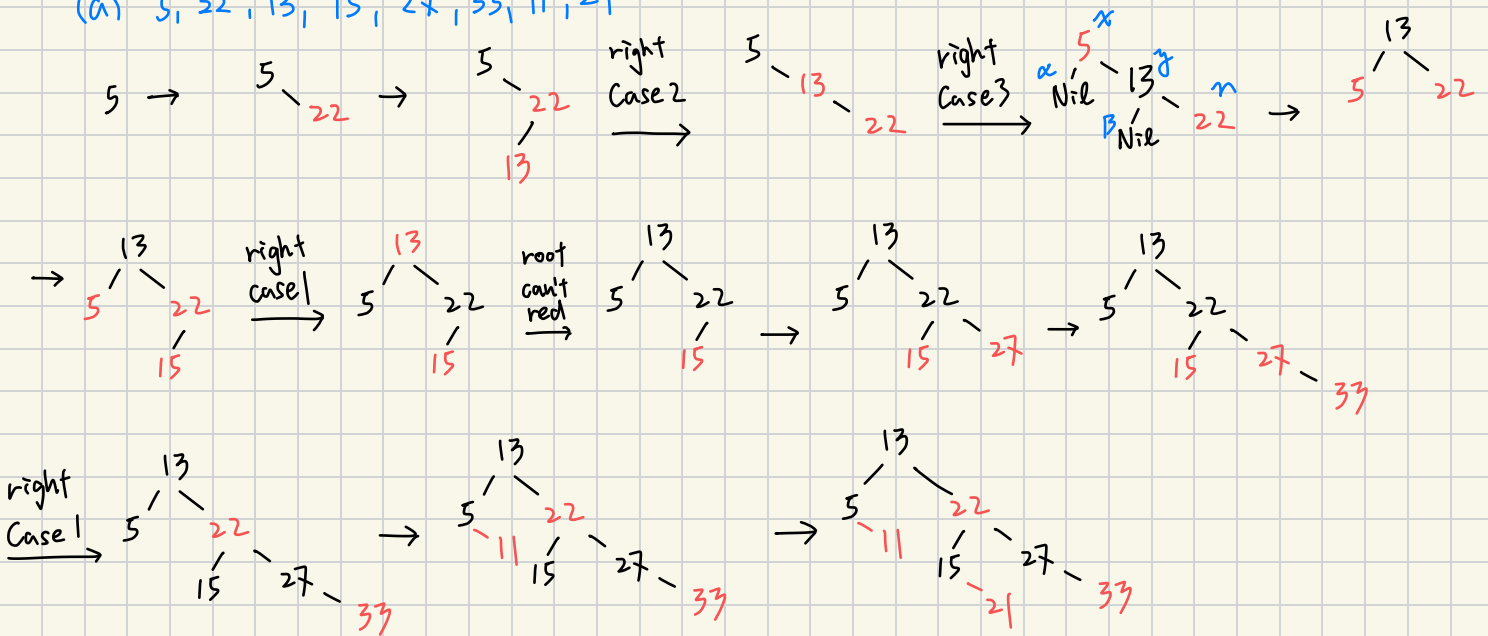
(d) $\langle 7, 27, 19, 15, 27, 23, 11, 26 \rangle$ $m=13$, linear hashing
 $\times \quad \times \quad \times \quad \times$

idx	0	1	2	3	4	5	6	7	8	9	10	11	12
val	26	27	15	27			19	7			23	11	

$\uparrow$ (at index 1)
 $\uparrow$ (at index 2)
 $\uparrow$ (at index 3)

3.

(a) $5, 22, 13, 15, 27, 33, 11, 21$



(b)

smallest number of internal nodes: $2^k - 1$
 largest number of internal nodes: $2^{2k} - 1$